

KRAKEN: Program-Adaptive Parallel Fuzzing (Supplement Material)

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1 BUG DETECTION RESULTS FOR REAL-WORLD PROGRAMS

We list the bug detection results for real-world programs in Table 1, including versions of applications, number of detected bugs, and number of CVEs assigned. We do not list all the CVE IDs during submission for the purpose of anonymity.

2 COMPLETE DATA FOR CORE NUMBER CHANGES

We show the complete results of parallelism degree (number of used CPU cores) changes during fuzzing all projects used in Section 4.2 of the original paper. The results are shown in Figure 1 and Figure 2.

3 COMPLETE DATA FOR EDGE COVERAGE CHANGES

We list the complete data for edge coverage changes within 3 hours using 10, 20, 30, and 40 CPU cores on all 12 programs in Figure 3 and Figure 4. The complete data for edge coverage changes within 10 hours using 20 cores on all 12 programs is shown in Figure 5.

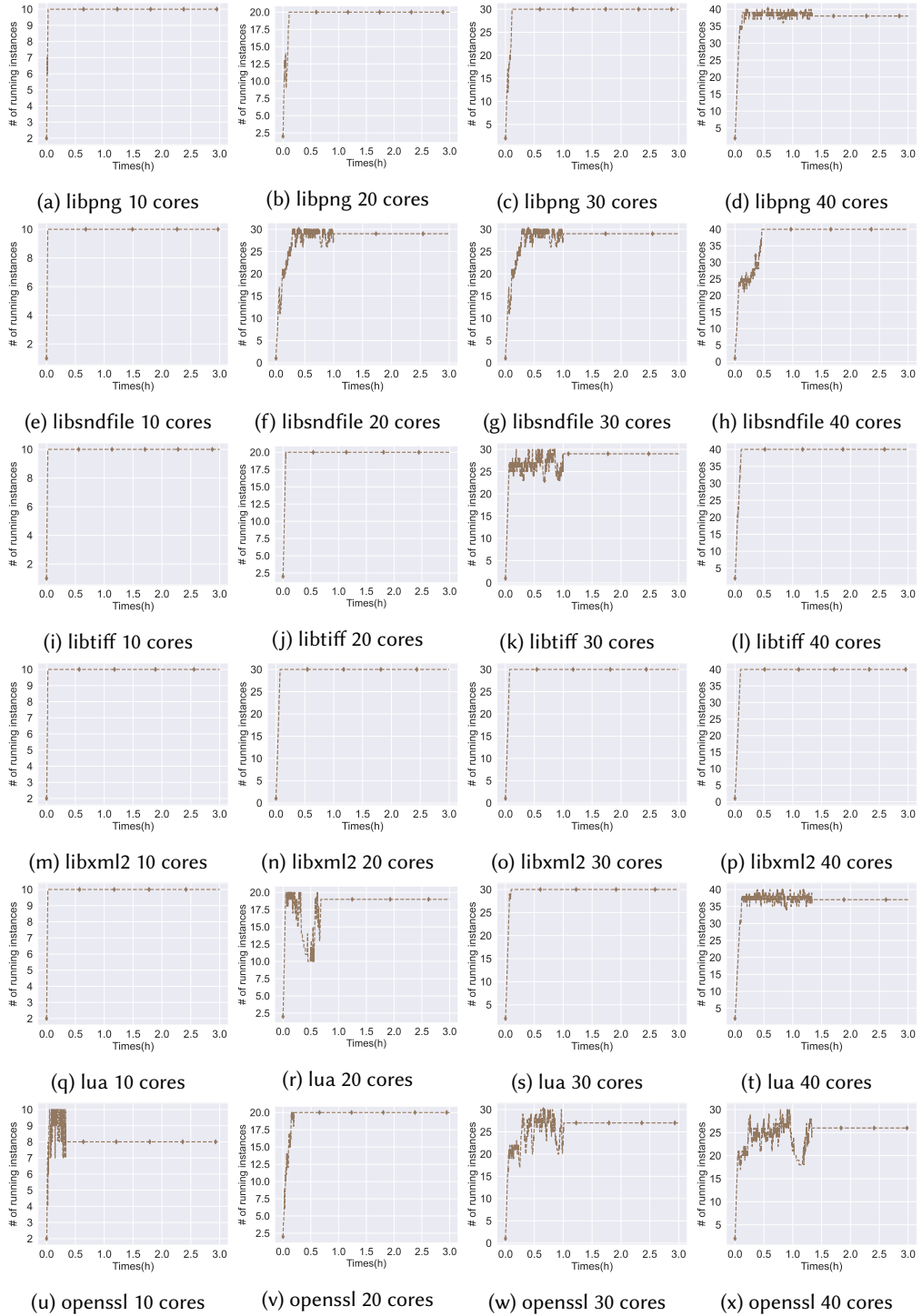


Fig. 1. The change in the number of running instances with time (3 hours) using 10, 20, 30 and 40 CPU cores.

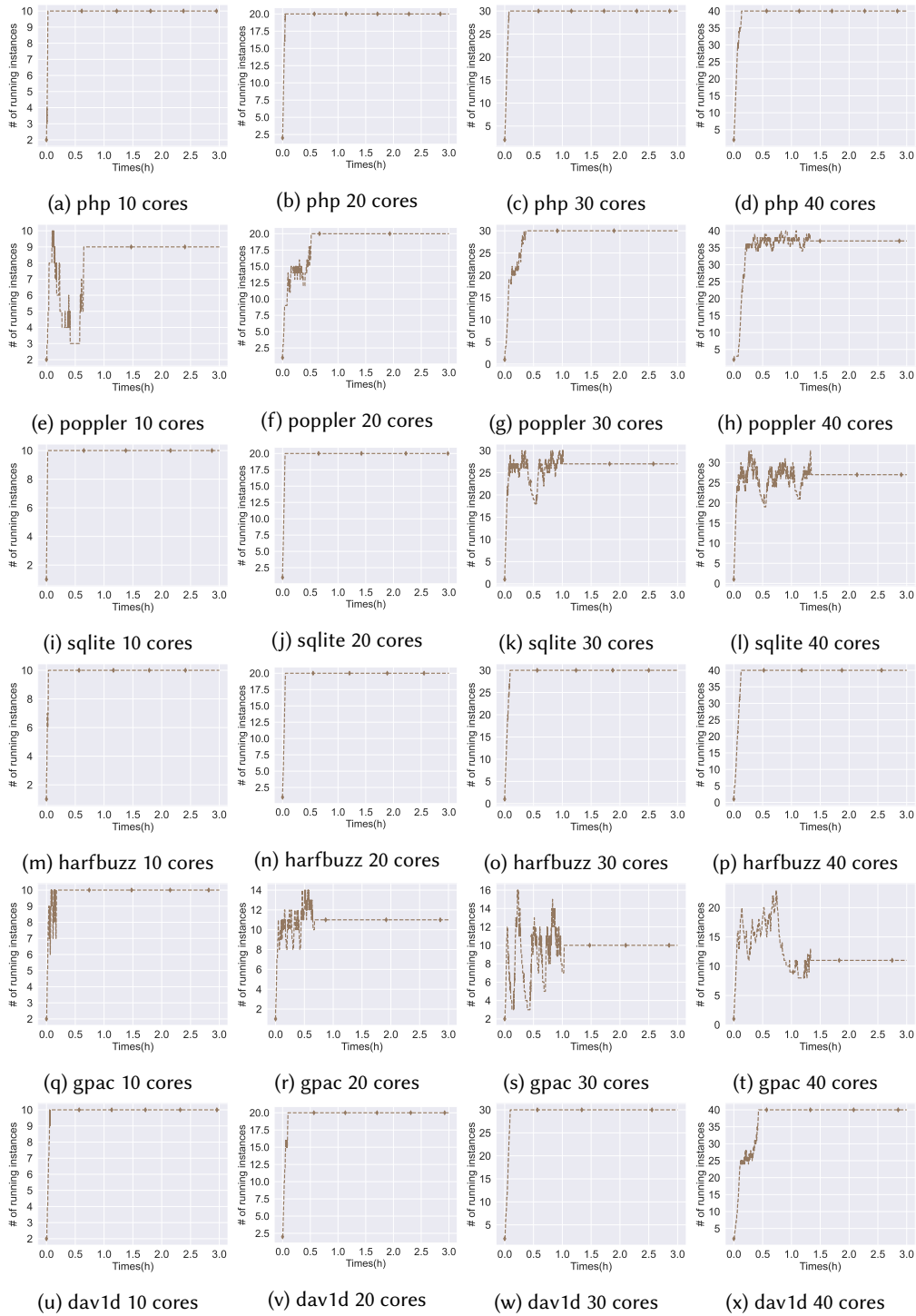


Fig. 2. The change in the number of running instances with time (3 hours) using 10, 20, 30 and 40 CPU cores (Cont.).

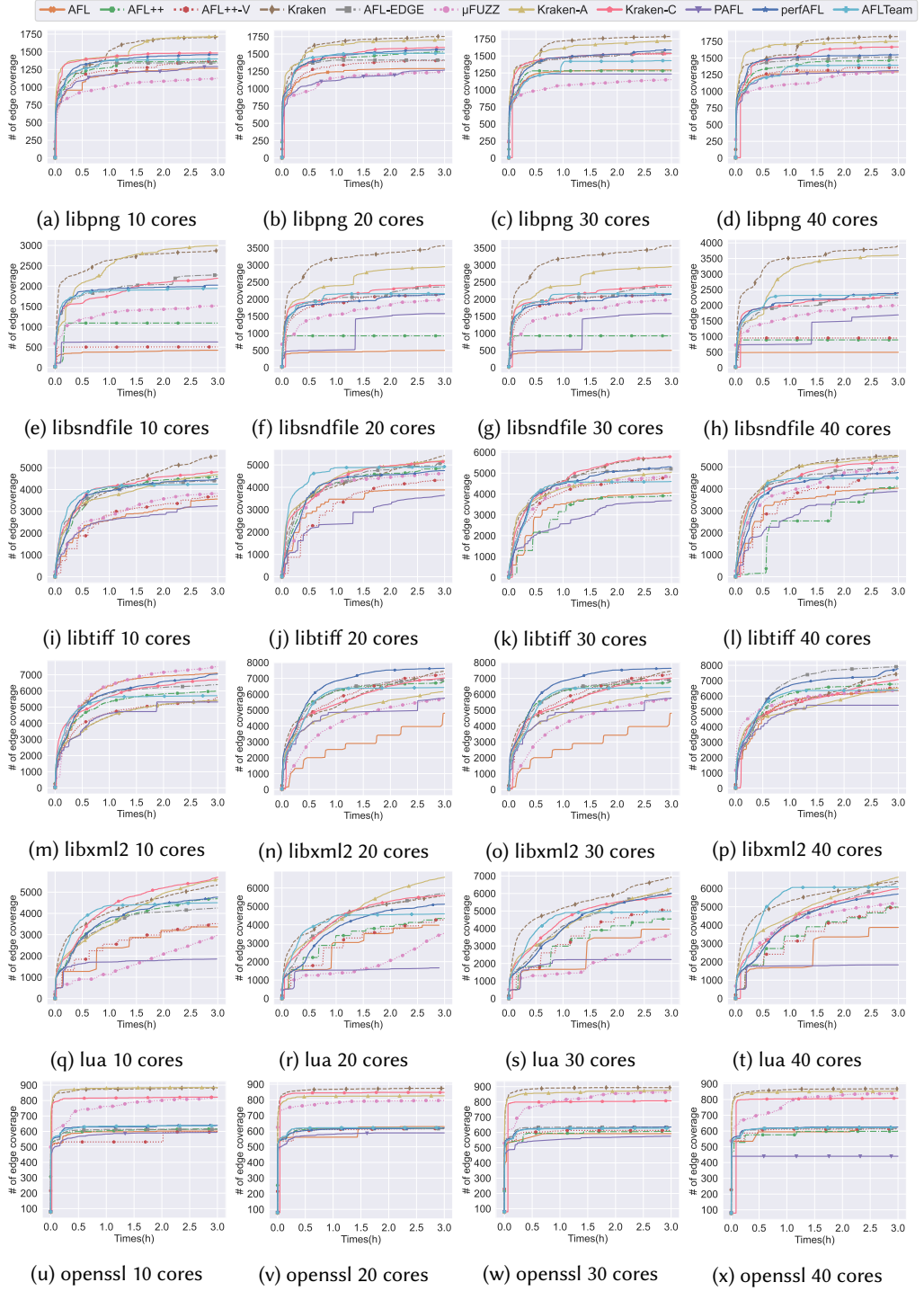


Fig. 3. Edge coverage changes with time (3 hours) using 10, 20, 30 and 40 CPU cores.

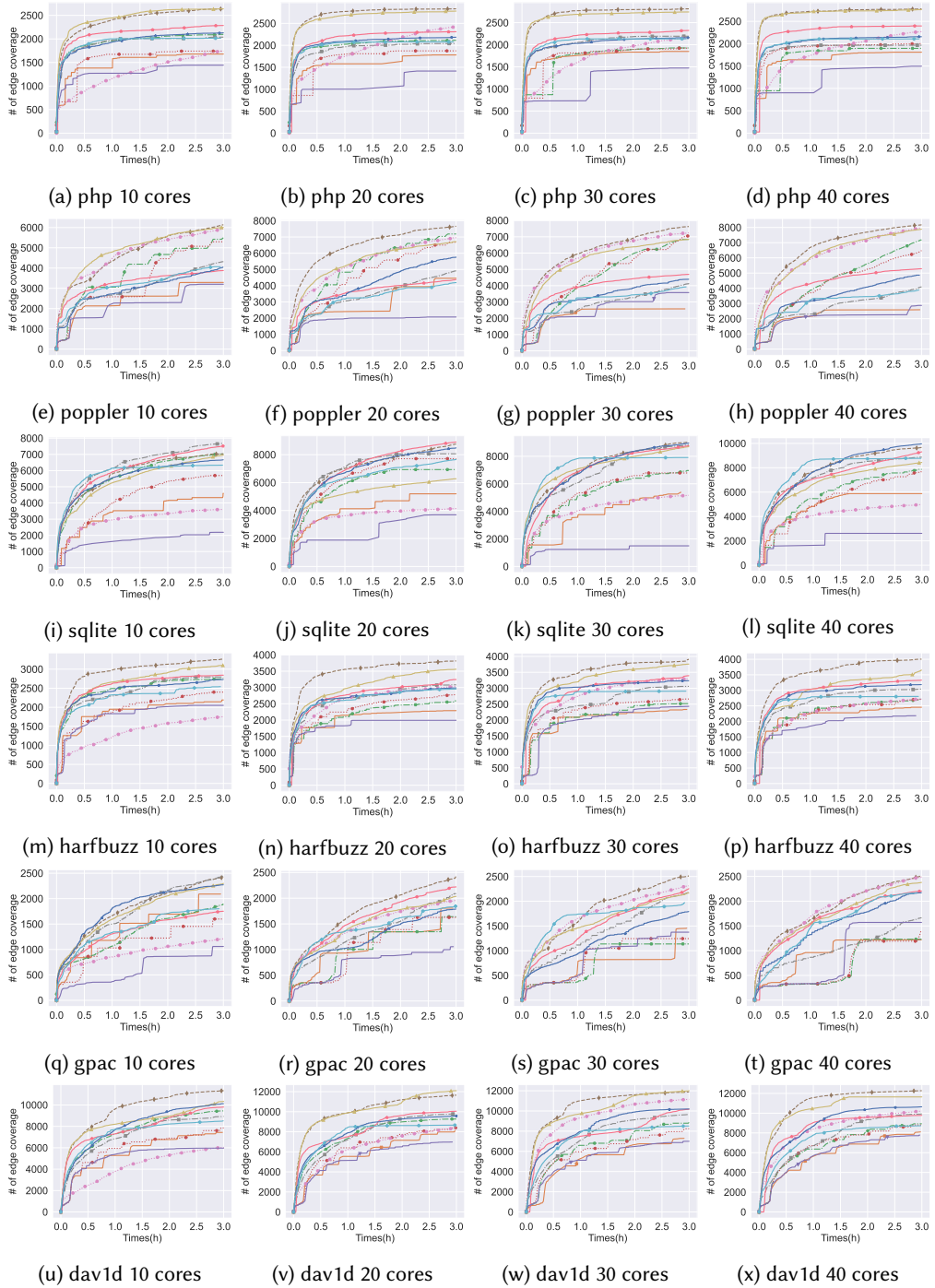


Fig. 4. Edge coverage changes with time (3 hours) using 10, 20, 30 and 40 CPU cores (Cont.).

Table 1. Detailed results of bug detection in real-world.

Application	Version	# of Bugs	# of CVEs
advancecomp	2.1	3	0
asn1c	0.9.28	4	0
Bento4	1.6.0	6	4
cpp-peglib	0.1.12	2	2
dmg2img	20170502	7	2
faad2	2.10.0	6	6
fast_ber	0.4	1	1
faust	2.30.5	1	1
ffmpeg	n4.5-dev	3	1
fig2dev	20200520	1	1
giflib	5.1.4	1	1
gocr	0.53	4	3
gpac	20200801	12	9
gravity	0.8.1	7	5
hcxtools	6.1.6	1	1
heif	3.6.2	4	3
jhead	1:3.04-2	2	0
jsonc	0.15	1	1
libelfin	20200729	1	1
libgig	20200507	1	1
libiff	20190123	2	1
libjpeg	2020021	19	7
libmaxminddb	1.4.2	1	1
libnsfdb	20200714	3	0
libraw	0.20.2	3	1
libredwg	0.10.1.3751	12	9
libslax	v0.22.1	4	4
libxsmm	v1.16.1	5	2
lief	0.11.4	1	1
nasm	2.15	2	0
ncurses	6.2	2	1
pbrt	20200627	1	1
pdftools	20200714	6	6
retdec	3.3	1	1
sela	20200412	9	9
sox	14.4.2	3	0
swftools	20200710	50	32

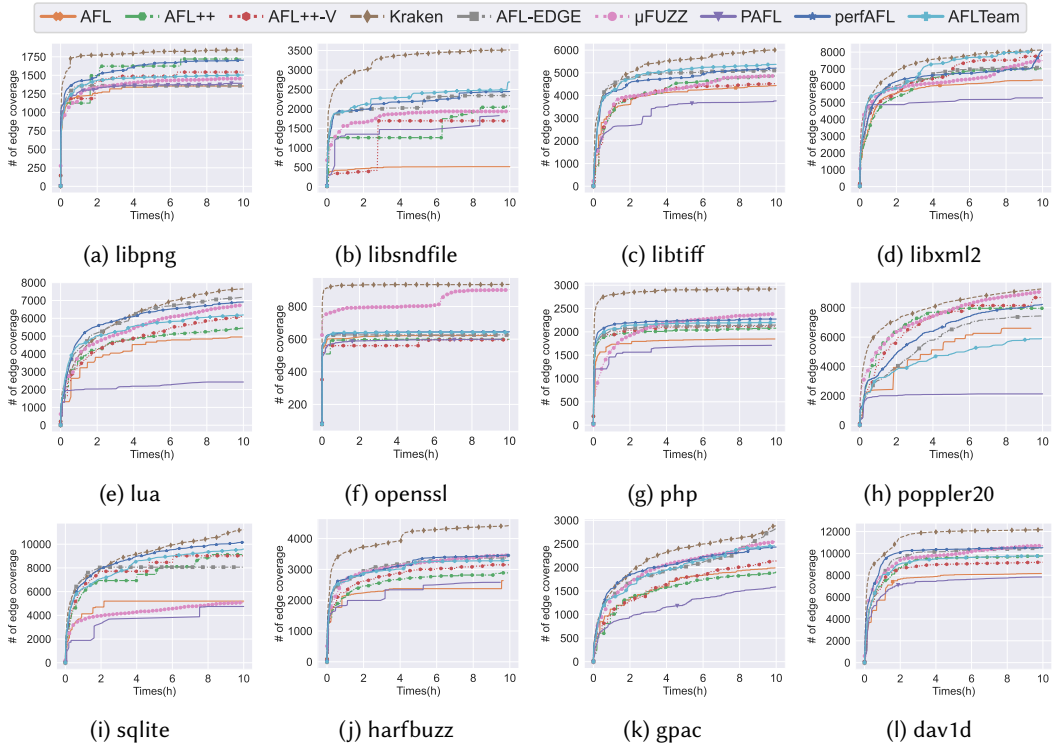


Fig. 5. Edge coverage changes in 10 hours using 20 cores.